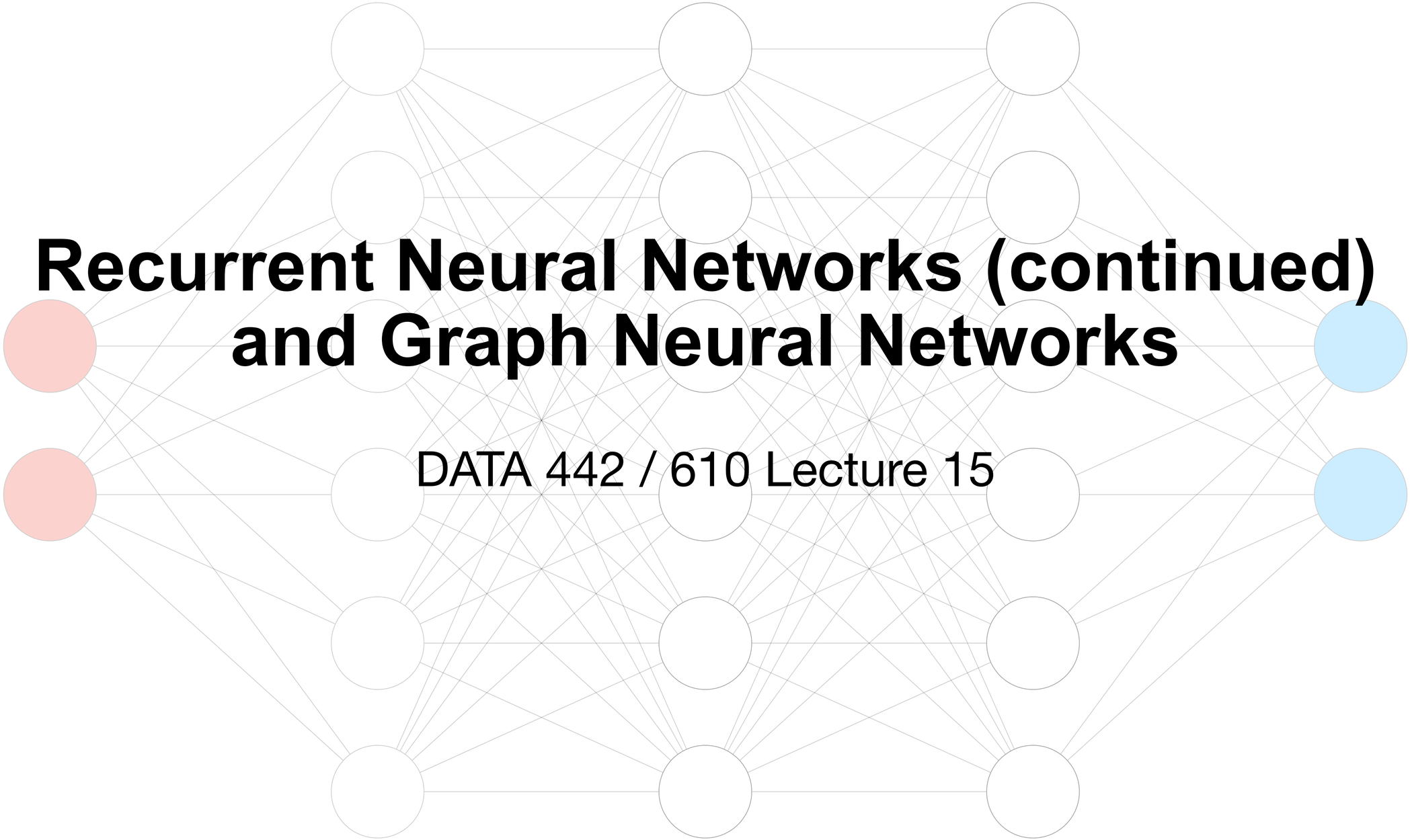


Recurrent Neural Networks (continued) and Graph Neural Networks

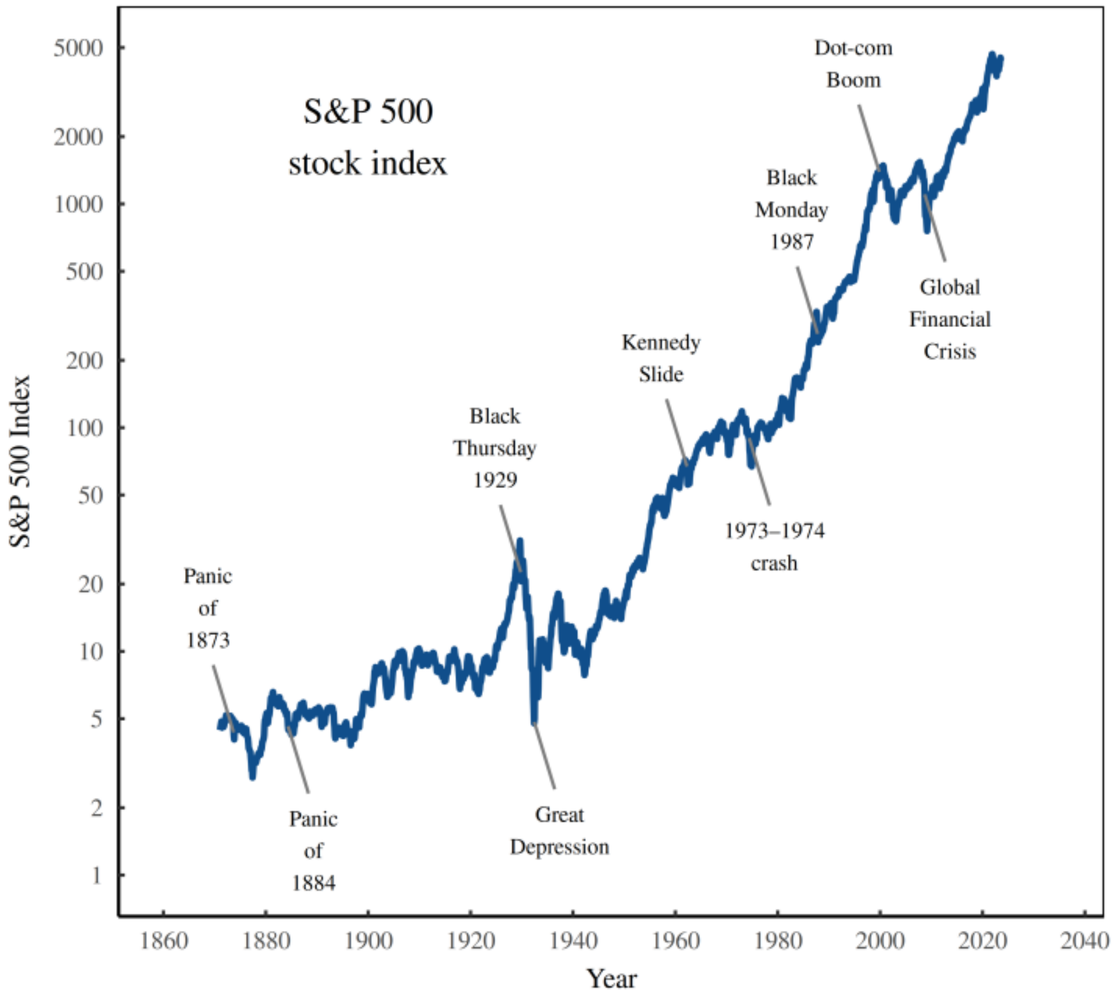
DATA 442 / 610 Lecture 15

Outline

- Time series forecasting
- Introduction to Graph Neural Networks

Reading: *Deep Learning*, Chapter 10 (RNNs), ArXiv:2104.13478 (GNNs)

Time-series forecasting



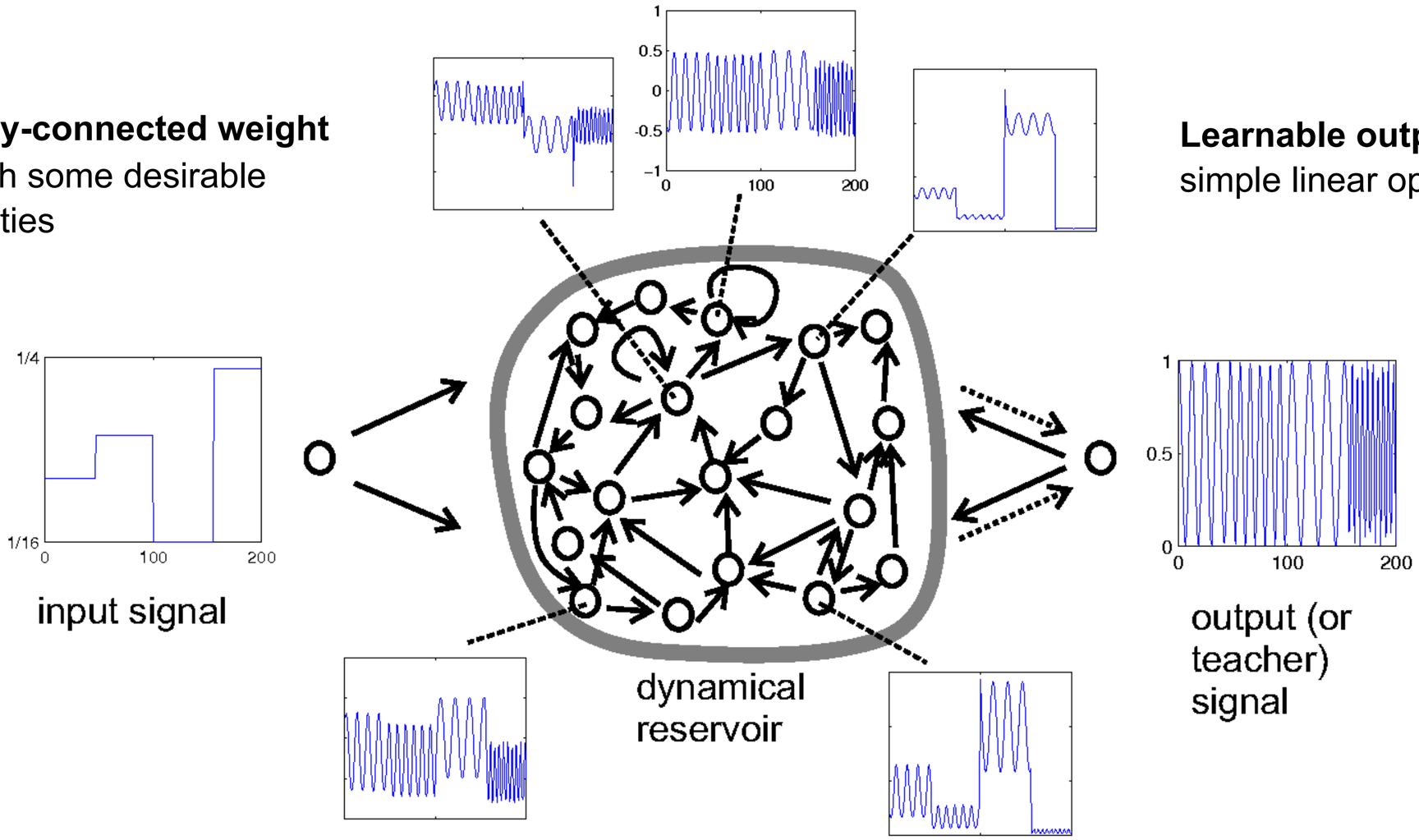


Hands-on with time-series forecasting



Reservoir computing and Echo State Networks

Fixed, sparsely-connected weight matrix W_{hh} with some desirable spectral properties



Learnable output matrix W_{oh} via simple linear optimization

https://en.wikipedia.org/wiki/Echo_state_network

A “liquid brain”

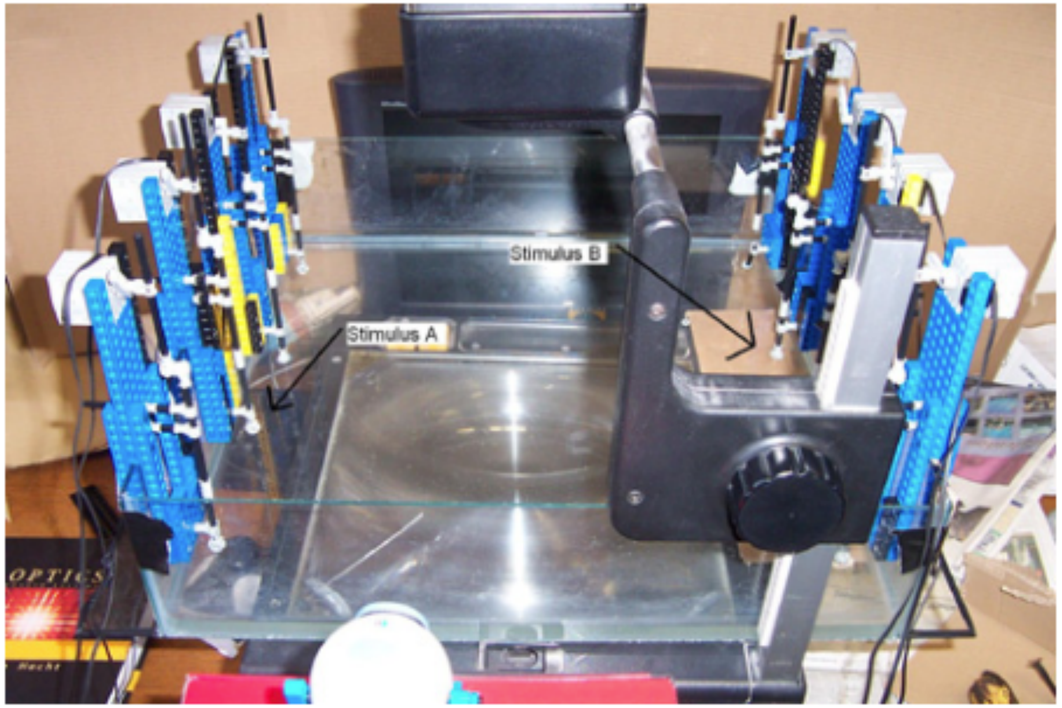


Fig. 1. The Liquid Brain.

Pattern Recognition in a Bucket

Chrisantha Fernando and Sampsa Sojakka

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Abstract. This paper demonstrates that the waves produced on the surface of water can be used as the medium for a “Liquid State Machine” that pre-processes inputs so allowing a simple perceptron to solve the XOR problem and undertake speech recognition. Interference between waves allows non-linear parallel computation upon simultaneous sensory inputs. Temporal patterns of stimulation are converted to spatial patterns of water waves upon which a linear discrimination can be made. Whereas Wolfgang Maass’ Liquid State Machine requires fine tuning of the spiking neural network parameters, water has inherent self-organising properties such as strong local interactions, time-dependent spread of activation to distant areas, inherent stability to a wide variety of inputs, and high complexity. Water achieves this “for free”, and does so without the time-consuming computation required by realistic neural models. An analogy is made between water molecules and neurons in a recurrent neural network.

More reservoir computing

Model-Free Prediction of Large Spatiotemporally Chaotic Systems from Data: A Reservoir Computing Approach

PDF

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Jaideep Pathak[1,2,*], Brian Hunt[3,4], Michelle Girvan[1,3,2], Zhixin Lu[1,3], and Edward Ott[1,2,5]

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Phys. Rev. Lett. **120**, 024102 – **Published 12 January, 2018**

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That’s all!

- Slides will be posted on the course website: https://colen-teaching.github.io/DATA442-Fall-2026
- Lab demos will also be posted
- ***PS2*** due October 25 - *note the extended due date*
- **REMEMBER:** TA Office Hours Tuesday 12-1pm, Student Success Center